Dominant Community State Types and Health Outcomes: Phase 1 Full-Cohort Analysis of the American Gut Project

Organized by Disease Outcome

Pawel Gajer

March 24, 2026

Abstract

We applied L_∞ normalization and Dominant Community State Type (DCST) classification to 30,290 samples from the American Gut Project (AGP) to investigate associations between gut microbiota community structure and 12 self-reported health conditions. Using two independent taxonomic schemas (SILVA 138 and GreenGenes2), we identified 40 depth-1 DCSTs and performed univariate (Fisher's exact test) and covariate-adjusted (logistic regression controlling for age, sex, BMI, and sequencing depth) association analyses. Sensitivity analysis excluding potentially oral/skin-contaminated samples assessed robustness. This report presents results organized by disease outcome, integrating findings across both taxonomies with published literature.

Contents

1 Methods **5**
- 1.1 Data and Cohort . . . 5
- 1.2 DCST Classification . . . 5
- 1.3 Statistical Analysis . . . 5
- 1.4 Covariates . . . 5

2 Cohort Overview **6**

3 Taxonomy Concordance: SILVA vs. GreenGenes2 **9**

4 Irritable Bowel Syndrome (IBS) **10**
- 4.1 Introduction . . . 10
- 4.2 Results . . . 10
  - 4.2.1 SILVA Univariate Associations . . . 10
  - 4.2.2 SILVA Covariate-Adjusted Associations . . . 10
  - 4.2.3 GreenGenes2 Replication . . . 10

4.2.4 Depth-2 Subtype Analysis . . . 10
4.2.5 Sensitivity Analysis . . . 10
4.3 Discussion . . . 11

5 Inflammatory Bowel Disease (IBD) 12
5.1 Introduction . . . 12
5.2 Results . . . 12
5.2.1 SILVA Univariate Associations . . . 12
5.2.2 SILVA Covariate-Adjusted Associations . . . 12
5.2.3 GreenGenes2 Replication . . . 12
5.2.4 Depth-2 Subtype Analysis . . . 12
5.2.5 Sensitivity Analysis . . . 13
5.3 Discussion . . . 13

6 Autoimmune Diseases 14
6.1 Introduction . . . 14
6.2 Results . . . 14
6.2.1 SILVA Univariate Associations . . . 14
6.2.2 SILVA Covariate-Adjusted Associations . . . 14
6.2.3 GreenGenes2 Replication . . . 14
6.2.4 Depth-2 Subtype Analysis . . . 14
6.2.5 Sensitivity Analysis . . . 14
6.3 Discussion . . . 15

7 Acid Reflux / GERD 16
7.1 Introduction . . . 16
7.2 Results . . . 16
7.2.1 SILVA Univariate Associations . . . 16
7.2.2 SILVA Covariate-Adjusted Associations . . . 16
7.2.3 GreenGenes2 Replication . . . 16
7.2.4 Depth-2 Subtype Analysis . . . 16
7.2.5 Sensitivity Analysis . . . 16
7.3 Discussion . . . 17

8 Cardiovascular Disease 18
8.1 Introduction . . . 18
8.2 Results . . . 18
8.2.1 SILVA Univariate Associations . . . 18
8.2.2 SILVA Covariate-Adjusted Associations . . . 18
8.2.3 GreenGenes2 Replication . . . 18
8.2.4 Depth-2 Subtype Analysis . . . 18
8.2.5 Sensitivity Analysis . . . 18
8.3 Discussion . . . 18

9 Obesity **20**
9.1 Introduction . . . 20
9.2 Results . . . 20
9.2.1 SILVA Univariate Associations . . . 20
9.2.2 SILVA Covariate-Adjusted Associations . . . 20
9.2.3 GreenGenes2 Replication . . . 20
9.2.4 Depth-2 Subtype Analysis . . . 20
9.2.5 Sensitivity Analysis . . . 20
9.3 Discussion . . . 20

10 Seasonal Allergies **22**
10.1 Introduction . . . 22
10.2 Results . . . 22
10.2.1 SILVA Univariate Associations . . . 22
10.2.2 SILVA Covariate-Adjusted Associations . . . 22
10.2.3 GreenGenes2 Replication . . . 22
10.2.4 Depth-2 Subtype Analysis . . . 22
10.2.5 Sensitivity Analysis . . . 22
10.3 Discussion . . . 22

11 Diabetes **24**
11.1 Introduction . . . 24
11.2 Results . . . 24
11.2.1 SILVA Univariate Associations . . . 24
11.2.2 SILVA Covariate-Adjusted Associations . . . 24
11.2.3 GreenGenes2 Replication . . . 24
11.2.4 Depth-2 Subtype Analysis . . . 24
11.2.5 Sensitivity Analysis . . . 24
11.3 Discussion . . . 24

12 *Clostridioides difficile* Infection (CDI) **25**
12.1 Introduction . . . 25
12.2 Results . . . 25
12.2.1 SILVA Univariate Associations . . . 25
12.2.2 SILVA Covariate-Adjusted Associations . . . 25
12.2.3 GreenGenes2 Replication . . . 25
12.2.4 Depth-2 Subtype Analysis . . . 25
12.2.5 Sensitivity Analysis . . . 25
12.3 Discussion . . . 25

13 Lung Disease **27**
13.1 Introduction . . . 27
13.2 Results . . . 27
13.2.1 SILVA Univariate Associations . . . 27

13.2.2 SILVA Covariate-Adjusted Associations . . . 27
13.2.3 GreenGenes2 Replication . . . 27
13.2.4 Depth-2 Subtype Analysis . . . 27
13.2.5 Sensitivity Analysis . . . 27
13.3 Discussion . . . 27

14 Migraine **29**
14.1 Introduction . . . 29
14.2 Results . . . 29
14.2.1 SILVA Univariate Associations . . . 29
14.2.2 SILVA Covariate-Adjusted Associations . . . 29
14.2.3 GreenGenes2 Replication . . . 29
14.2.4 Depth-2 Subtype Analysis . . . 29
14.2.5 Sensitivity Analysis . . . 29
14.3 Discussion . . . 29

15 Kidney Disease **30**
15.1 Introduction . . . 30
15.2 Results . . . 30
15.2.1 SILVA Univariate Associations . . . 30
15.2.2 SILVA Covariate-Adjusted Associations . . . 30
15.2.3 GreenGenes2 Replication . . . 30
15.2.4 Depth-2 Subtype Analysis . . . 30
15.2.5 Sensitivity Analysis . . . 30
15.3 Discussion . . . 30

16 General Discussion **32**
16.1 Limitations . . . 32
16.2 Conclusions . . . 32

1 Methods

1.1 Data and Cohort

We analyzed 16S rRNA gene amplicon sequencing data from 34,711 AGP samples McDonald et al. [2018]. After quality filtering (library size ≥ 1000 reads, > 0 non-zero taxa), 30,290 samples across 274 taxa (SILVA 138 genus level Quast et al. [2013]) were retained.

1.2 DCST Classification

Each sample was classified by its dominant taxon after L_∞ normalization (dividing each row by its maximum value). DCSTs were defined with a minimum size threshold $n_0 = 50$; samples whose dominant taxon fell below this threshold were aggregated into a `RARE_DOMINANT` category. This yielded 40 depth-1 DCSTs plus one rare-dominant group. Depth-2 refinement subdivided each depth-1 DCST by its second-most-abundant taxon, yielding 123 depth-2 subtypes. The concept of community state types based on dominant species follows Gajer et al. [2012].

1.3 Statistical Analysis

Univariate: Fisher's exact test for each DCST–condition pair, with Benjamini–Hochberg correction Benjamini and Hochberg [1995] at $q < 0.05$. **Adjusted:** Logistic regression with covariates: age (midpoint of AGP categorical bin), sex, BMI, and $\log_{10}$ sequencing depth. **Sensitivity:** Repeated adjusted analysis after excluding 2,229 samples dominated by putative oral or skin taxa (Streptococcus, Staphylococcus, Rothia, Corynebacterium, Neisseria, Veillonella). **GG2 replication:** Independent analysis using GreenGenes2 taxonomy McDonald et al. [2024] on the same samples. **Depth-2:** Adjusted analysis on 123 depth-2 subtypes.

1.4 Covariates

Age was available for 27,372 samples (90.4%), sex for 25,598 (84.5%), and BMI for 26,541 (87.6%). Complete cases for adjusted analysis: 22,632 (74.7%). Obesity was defined as BMI ≥ 30.

2 Cohort Overview

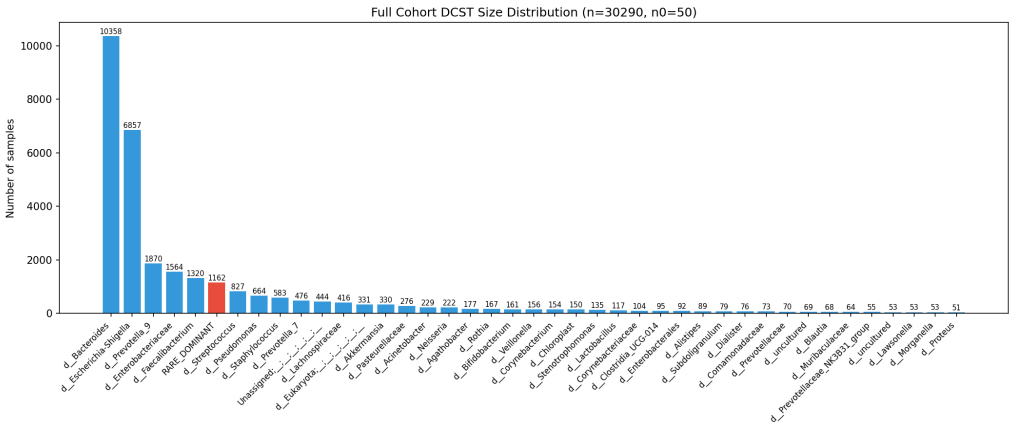

Figure 1: Distribution of depth-1 DCST sizes across 30,290 AGP samples (SILVA taxonomy). *Bacteroides* is the largest DCST (10,358 samples), followed by *Escherichia–Shigella* (6,857).

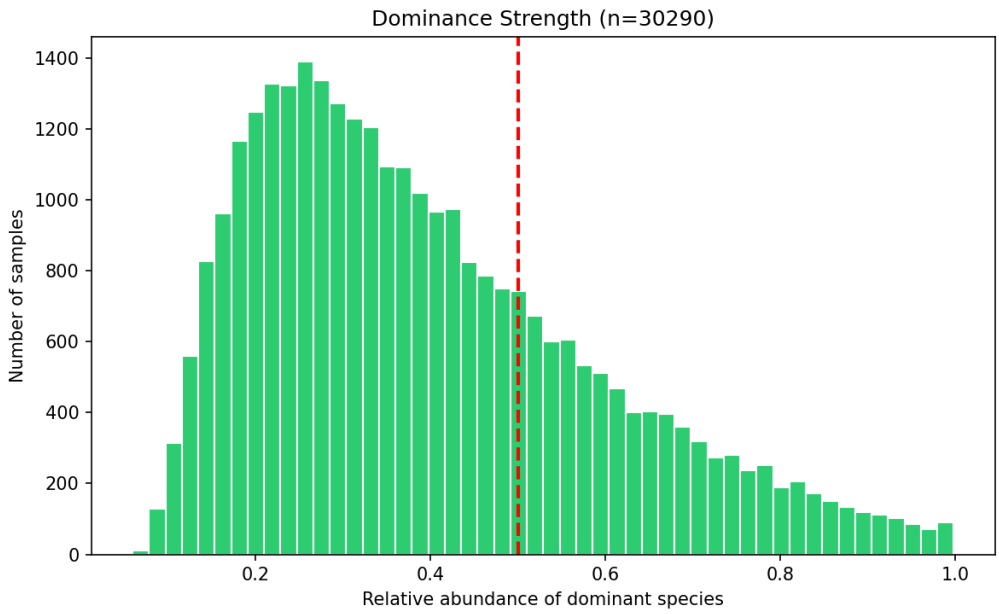

Figure 2: Distribution of dominance strength (L_∞-normalized maximum) across DCSTs. Higher values indicate stronger dominance by a single taxon.

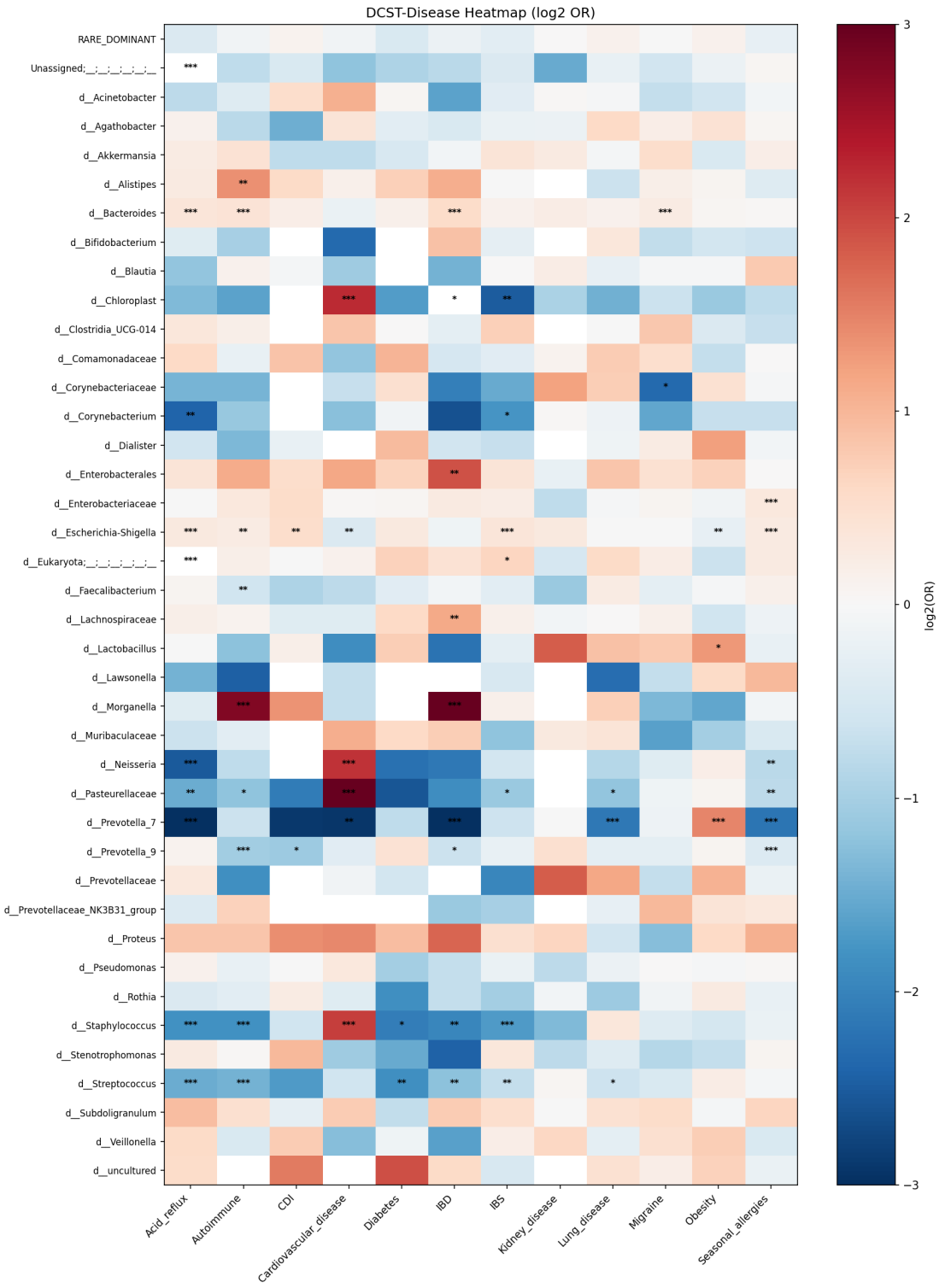

Figure 3: Heatmap of univariate DCST–disease association significance ($-\log_{10} q$). Red cells indicate enrichment (OR>1), blue indicate depletion (OR<1).

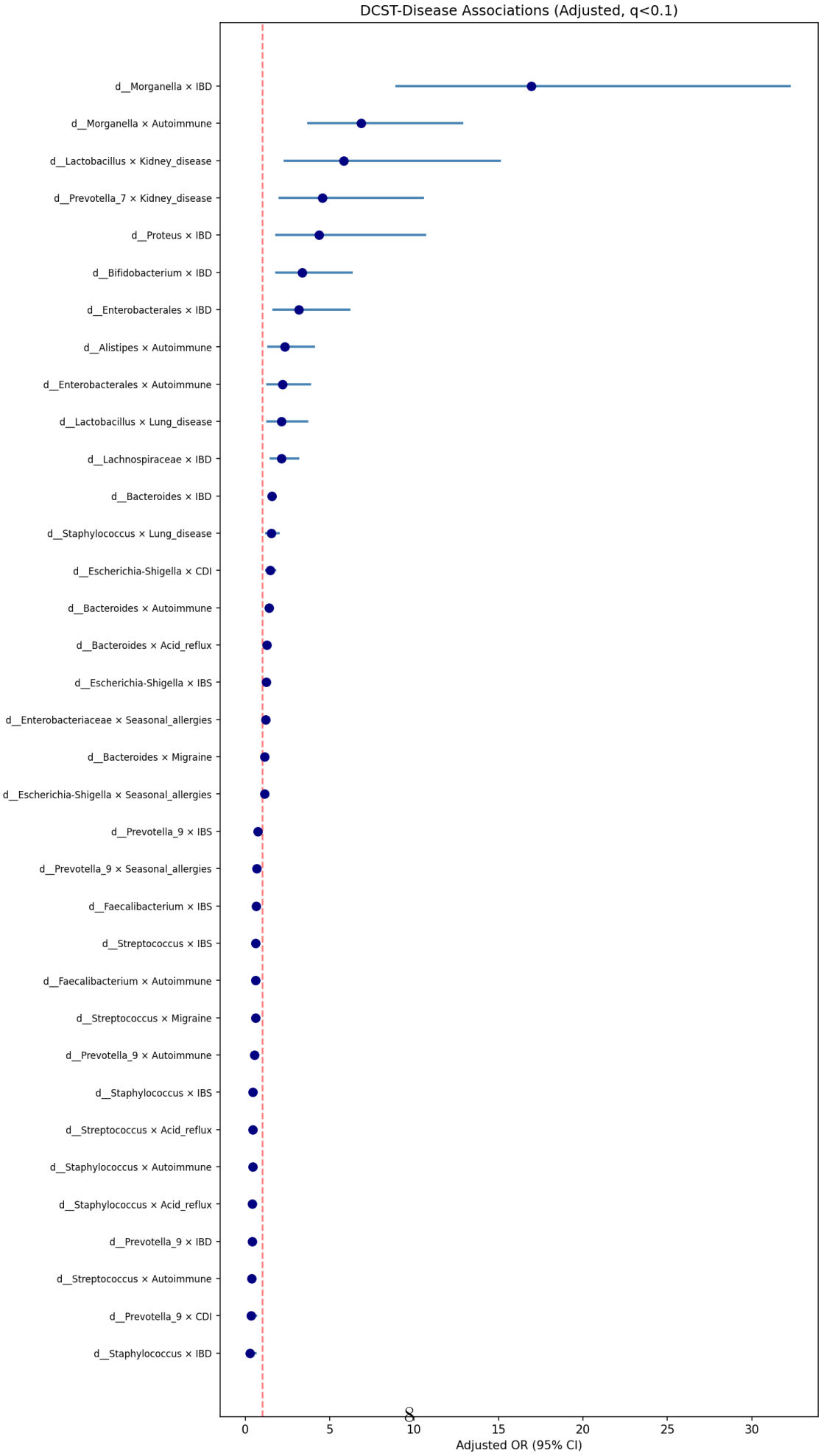

Figure 4: Forest plot of covariate-adjusted odds ratios for significant DCST–disease associations

3 Taxonomy Concordance: SILVA vs. GreenGenes2

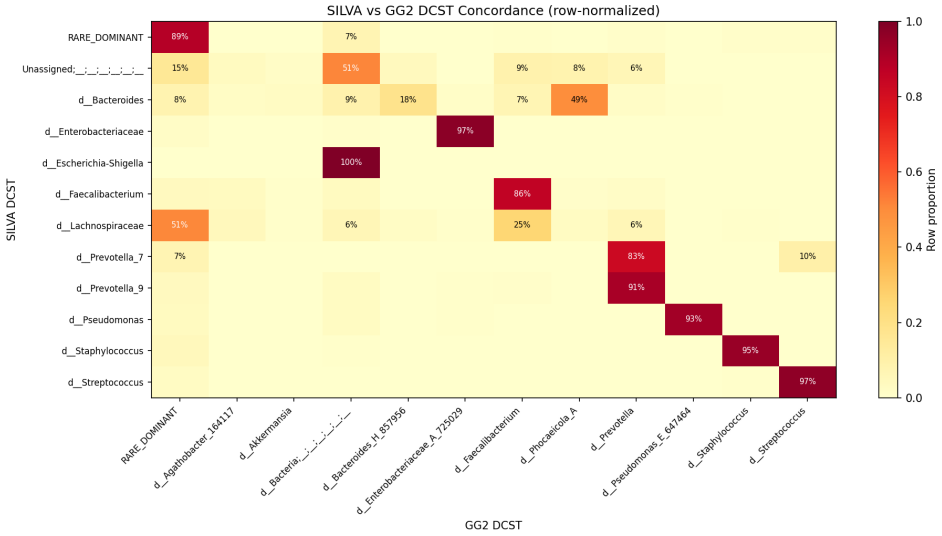

Figure 5: Concordance heatmap showing mapping between SILVA and GG2 dominant-taxon assignments. Overall genus-level concordance: 91.6%.

Independent DCST classification using GreenGenes2 McDonald et al. [2024] identified 59 depth-1 DCSTs. Key taxonomic mappings include: SILVA *Bacteroides* maps predominantly to GG2 *Phocaeicola_A* (47%) and *Bacteroides_H* (34%); SILVA *Escherichia–Shigella* maps 100% to GG2 "Bacteria;___" (unresolved); *Akkermansia* shows 99% concordance. Overall, 91.6% of samples received concordant genus-level dominant-taxon assignments.

4 Irritable Bowel Syndrome (IBS)

4.1 Introduction

Irritable bowel syndrome (IBS) is a functional gastrointestinal disorder affecting 10–15% of the global population, characterized by chronic abdominal pain, bloating, and altered bowel habits Pittayanon et al. [2019]. The gut microbiome has emerged as a central factor in IBS pathophysiology through the microbiota–gut–brain axis Dinan and Cryan [2015], Cryan and Dinan [2012]. Systematic reviews identify consistent compositional shifts including increased Enterobacteriaceae and *Bacteroides*, decreased *Faecalibacterium prausnitzii* and *Bifidobacterium*, and reduced overall diversity Pittayanon et al. [2019]. A 90-OTU microbiota signature has been linked to IBS severity Tap et al. [2017]. Here we present DCST-based associations with IBS in the American Gut Project (AGP) cohort.

4.2 Results

4.2.1 SILVA Univariate Associations

Significant univariate DCST associations with irritable bowel syndrome: *Staphylococcus* (depleted, OR=0.31, $q = 2.2e-08$), *Escherichia-Shigella* (enriched, OR=1.29, $q = 7.8e-08$), *Streptococcus* (depleted, OR=0.60, $q = 2.6e-03$), *Chloroplast* (depleted, OR=0.18, $q = 2.7e-03$), *Corynebacterium* (depleted, OR=0.29, $q = 1.9e-02$), *Pasteurellaceae* (depleted, OR=0.47, $q = 2.5e-02$), *Eukaryota; ; ; ; ; ;* (enriched, OR=1.57, $q = 5.0e-02$).

4.2.2 SILVA Covariate-Adjusted Associations

Table 1: Covariate-adjusted DCST associations with Irritable Bowel Syndrome (IBS) (SILVA, $q < 0.05$).

DCST	adjOR	95% CI	q-value
Escherichia-Shigella	1.23	1.12–1.35	6.8e-04
Faecalibacterium	0.64	0.50–0.82	9.5e-03
Staphylococcus	0.46	0.29–0.73	2.0e-02
Streptococcus	0.61	0.44–0.85	5.0e-02

4.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Bacteria; ; ; ; ; ;* (enriched, OR=1.30), *Staphylococcus* (depleted, OR=0.32), *Streptococcus* (depleted, OR=0.57), *Corynebacterium* (depleted, OR=0.34), *Bacteroides H 857956* (enriched, OR=2.11) (showing top 5 of 7).

4.2.4 Depth-2 Subtype Analysis

Escherichia-Shigella (adjOR=1.55, $q = 3.2e-04$), *Bacteroides* (adjOR=1.23, $q = 3.7e-02$).

4.2.5 Sensitivity Analysis

3 association(s) survived: *Faecalibacterium* (adjOR=0.62), *Escherichia-Shigella* (adjOR=1.19), *Prevotella 9* (adjOR=0.71).

4.3 Discussion

Our finding of enriched *Escherichia–Shigella*-dominant DCSTs in IBS aligns with the well-documented expansion of Enterobacteriaceae in IBS Pittayanon et al. [2019]. The protective association of *Faecalibacterium*-dominant DCSTs is consistent with the anti-inflammatory properties of *F. prausnitzii* and its depletion in IBS patients Sokol et al. [2008]. The reduced prevalence of *Streptococcus* and *Staphylococcus* DCSTs among IBS subjects is notable; these oral-associated taxa may reflect sample characteristics rather than true protective effects, though the association persisted in the sensitivity analysis excluding potential oral contaminants for *Faecalibacterium*. The microbiota–gut–brain axis framework suggests that these compositional shifts affect visceral sensitivity, motility, and immune activation through altered SCFA production and serotonin signaling Dinan and Cryan [2015], Cryan and Dinan [2012].

5 Inflammatory Bowel Disease (IBD)

5.1 Introduction

Inflammatory bowel disease encompasses Crohn's disease and ulcerative colitis and affects over 6 million people worldwide. The gut microbiome plays a fundamental role in IBD pathogenesis, with landmark studies demonstrating reduced microbial diversity, expansion of Proteobacteria, and depletion of butyrate-producing Firmicutes Lloyd-Price et al. [2019], Gevers et al. [2014], Frank et al. [2007]. The iHMP longitudinal study confirmed that not only composition but also microbial transcription and metabolite profiles are disrupted during disease flares Lloyd-Price et al. [2019]. Depletion of *Faecalibacterium prausnitzii*, a potent anti-inflammatory commensal, is a hallmark of Crohn's disease Sokol et al. [2008], Manichanh et al. [2006].

5.2 Results

5.2.1 SILVA Univariate Associations

Significant univariate DCST associations with inflammatory bowel disease: *Morganella* (enriched, OR=18.05, $q = 9.1e-16$), *Bacteroides* (enriched, OR=1.46, $q = 2.2e-08$), *Prevotella 7* (depleted, OR=0.10, $q = 5.8e-05$), *Staphylococcus* (depleted, OR=0.26, $q = 1.2e-03$), *Lachnospiraceae* (enriched, OR=2.19, $q = 1.5e-03$), *Enterobacterales* (enriched, OR=3.77, $q = 2.8e-03$), *Streptococcus* (depleted, OR=0.42, $q = 5.3e-03$), *Prevotella 9* (depleted, OR=0.64, $q = 2.2e-02$), *Chloroplast* (depleted, OR=0.00, $q = 4.1e-02$).

5.2.2 SILVA Covariate-Adjusted Associations

Table 2: Covariate-adjusted DCST associations with Inflammatory Bowel Disease (IBD) (SILVA, $q < 0.05$).

DCST	adjOR	95% CI	q-value
Morganella	16.94	8.88–32.31	4.0e-15
Bacteroides	1.56	1.37–1.78	3.5e-09
Prevotella 9	0.41	0.27–0.61	5.5e-04
Lachnospiraceae	2.14	1.43–3.19	6.2e-03
Bifidobacterium	3.36	1.77–6.37	6.2e-03
Enterobacterales	3.15	1.60–6.24	2.0e-02
Proteus	4.37	1.78–10.73	2.4e-02

5.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Lachnospira* (enriched, OR=116.44), *Morganella* (enriched, OR=17.70), *Alistipes A 871400* (enriched, OR=11.63), *Streptococcus* (depleted, OR=0.28), *Proteus* (enriched, OR=5.52) (showing top 5 of 10).

5.2.4 Depth-2 Subtype Analysis

Bacteroides (adjOR=3.54, $q = 1.6e-37$), *Morganella* (adjOR=16.94, $q = 4.0e-15$), *Bacteroides* (adjOR=4.04, $q = 2.1e-04$).

5.2.5 Sensitivity Analysis

8 association(s) survived: *Morganella* (adjOR=16.33), *Bacteroides* (adjOR=1.48), *Prevotella 9* (adjOR=0.39), *Bifidobacterium* (adjOR=3.21), *Lachnospiraceae* (adjOR=2.06), *Enterobacterales* (adjOR=3.02), *Escherichia-Shigella* (adjOR=0.78), *Proteus* (adjOR=4.20).

5.3 Discussion

The striking association between *Morganella*-dominant DCSTs and IBD (adjOR=16.94) represents one of the strongest signals in our analysis. *Morganella morganii*, a member of Enterobacteriaceae, is an opportunistic pathogen whose expansion in the inflamed gut is consistent with the oxygen hypothesis: mucosal inflammation generates reactive oxygen species that favor facultative anaerobes over obligate anaerobes Sartor [2008], Frank et al. [2007]. The enrichment of *Bacteroides*-dominant DCSTs (adjOR=1.56) aligns with observations from the iHMP that *Bacteroides* species expand during IBD flares, potentially through exploitation of host-derived glycans Lloyd-Price et al. [2019]. Conversely, the protective *Prevotella_9* association (adjOR=0.41) mirrors the ecological incompatibility between Prevotella-dominated and Bacteroides-dominated community states Arumugam et al. [2011]. The *Lachnospiraceae* and *Bifidobacterium* DCST enrichments in IBD may reflect compensatory community restructuring after loss of other commensals. These associations were robust across taxonomic schemas (SILVA and GG2) and survived sensitivity analysis, supporting their biological relevance Gevers et al. [2014].

6 Autoimmune Diseases

6.1 Introduction

Autoimmune diseases arise from aberrant immune responses against self-antigens, and the gut microbiome has emerged as a key modulator of immune tolerance. Expansion of *Prevotella copri* in rheumatoid arthritis Scher et al. [2013], depletion of regulatory T-cell-inducing Clostridia Atarashi et al. [2013], and disease-specific dysbiosis across SLE, multiple sclerosis, and type 1 diabetes have been documented De Luca and Shoenfeld [2019], Vatanen et al. [2018]. Mechanistic pathways include molecular mimicry, impaired SCFA-mediated Treg induction, and barrier dysfunction permitting bacterial translocation.

6.2 Results

6.2.1 SILVA Univariate Associations

Significant univariate DCST associations with autoimmune diseases: *Prevotella 9* (depleted, OR=0.49, $q = 1.6e-11$), *Bacteroides* (enriched, OR=1.34, $q = 4.5e-11$), *Staphylococcus* (depleted, OR=0.28, $q = 1.8e-08$), *Morganella* (enriched, OR=6.82, $q = 2.2e-08$), *Streptococcus* (depleted, OR=0.38, $q = 2.2e-08$), *Escherichia-Shigella* (enriched, OR=1.19, $q = 2.0e-03$), *Faecalibacterium* (depleted, OR=0.66, $q = 2.1e-03$), *Alistipes* (enriched, OR=2.60, $q = 7.8e-03$), *Pasteurellaceae* (depleted, OR=0.43, $q = 1.9e-02$).

6.2.2 SILVA Covariate-Adjusted Associations

Table 3: Covariate-adjusted DCST associations with Autoimmune Diseases (SILVA, $q < 0.05$).

DCST	adjOR	95% CI	q-value
Bacteroides	1.39	1.28–1.52	2.1e-11
Morganella	6.88	3.66–12.92	1.9e-07
Prevotella 9	0.53	0.42–0.68	2.6e-05
Streptococcus	0.36	0.23–0.56	2.0e-04
Faecalibacterium	0.61	0.47–0.79	5.9e-03
Staphylococcus	0.43	0.26–0.71	2.0e-02

6.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Lachnospira* (enriched, OR=46.76), *Morganella* (enriched, OR=7.15), *Streptococcus* (depleted, OR=0.35), *Staphylococcus* (depleted, OR=0.29), *Prevotella* (depleted, OR=0.48) (showing top 5 of 16).

6.2.4 Depth-2 Subtype Analysis

Bacteroides (adjOR=2.25, $q = 1.2e-23$), *Morganella* (adjOR=6.88, $q = 7.0e-07$), *Bacteroides* (adjOR=1.89, $q = 8.2e-07$).

6.2.5 Sensitivity Analysis

4 association(s) survived: *Bacteroides* (adjOR=1.33), *Morganella* (adjOR=6.63), *Prevotella 9* (adjOR=0.51), *Faecalibacterium* (adjOR=0.59).

6.3 Discussion

The *Bacteroides*-dominant DCST enrichment in autoimmune disease (adjOR=1.39) resonates with literature documenting *Bacteroides fragilis* as both protective (via polysaccharide A induction of IL-10-producing Tregs) and pathogenic (enterotoxigenic strains promoting Th17 responses) depending on strain context De Luca and Shoenfeld [2019]. The strong *Morganella* association (adjOR=6.88) parallels the IBD findings and likely reflects shared inflammatory mechanisms. The protective *Prevotella_9* (adjOR=0.53), *Faecalibacterium* (adjOR=0.61), and *Streptococcus* associations align with the framework of SCFA-producing commensals maintaining immune tolerance through Treg induction Atarashi et al. [2013]. These patterns were confirmed with GG2 taxonomy and survived sensitivity analysis, with *Bacteroides*, *Morganella*, *Prevotella_9*, and *Faecalibacterium* all remaining significant.

7 Acid Reflux / GERD

7.1 Introduction

Gastroesophageal reflux disease (GERD) affects approximately 20% of the Western population. Recent research has shifted the understanding of GERD pathophysiology from a purely acid-mediated mechanism to include microbial dysbiosis D'Souza et al. [2021]. Two distinct esophageal microbiome phenotypes have been identified: a healthy Streptococcus-dominated type and a GERD-associated gram-negative-enriched type Deshpande et al. [2018]. Mendelian randomization studies have identified causal associations between specific gut microbial taxa and GERD risk Wang et al. [2024].

7.2 Results

7.2.1 SILVA Univariate Associations

Significant univariate DCST associations with acid reflux / gerd: *Unassigned; ; ; ; ; ;* (depleted, OR=0.00, $q = 1.8e-17$), *Eukaryota; ; ; ; ; ;* (depleted, OR=0.00, $q = 4.9e-13$), *Prevotella 7* (depleted, OR=0.12, $q = 5.7e-12$), *Streptococcus* (depleted, OR=0.35, $q = 4.5e-09$), *Bacteroides* (enriched, OR=1.29, $q = 9.6e-09$), *Staphylococcus* (depleted, OR=0.28, $q = 2.2e-08$), *Neisseria* (depleted, OR=0.17, $q = 1.5e-04$), *Escherichia-Shigella* (enriched, OR=1.22, $q = 2.0e-04$), *Pasteurellaceae* (depleted, OR=0.36, $q = 4.7e-03$), *Corynebacterium* (depleted, OR=0.19, $q = 5.1e-03$).

7.2.2 SILVA Covariate-Adjusted Associations

Table 4: Covariate-adjusted DCST associations with Acid Reflux / GERD (SILVA, $q < 0.05$).

DCST	adjOR	95% CI	q-value
Bacteroides	1.28	1.17–1.40	6.9e-06
Streptococcus	0.45	0.30–0.67	4.2e-03
Staphylococcus	0.41	0.25–0.68	1.1e-02

7.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Streptococcus* (depleted, OR=0.30), *Prevotella* (depleted, OR=0.13), *Staphylococcus* (depleted, OR=0.33), *Neisseria 563205* (depleted, OR=0.18), *Pseudomonadaceae* (enriched, OR=2.43) (showing top 5 of 9).

7.2.4 Depth-2 Subtype Analysis

Parabacteroides (adjOR=1.99, $q = 6.6e-04$), *Alistipes* (adjOR=1.51, $q = 3.7e-02$).

7.2.5 Sensitivity Analysis

1 association(s) survived: *Bacteroides* (adjOR=1.21).

7.3 Discussion

The enrichment of *Bacteroides*-dominant DCSTs in acid reflux (adjOR=1.28) and protective associations for *Streptococcus* and *Staphylococcus* DCSTs are noteworthy. The latter may reflect the observation that healthy esophageal microbiomes are dominated by gram-positive organisms Deshpande et al. [2018], and individuals whose gut is dominated by these taxa may represent a healthier baseline. The protective *Streptococcus* association survived covariate adjustment but only *Bacteroides* survived the full sensitivity analysis, suggesting potential confounding for some associations. The strong univariate depletion of Unassigned and Eukaryota DCSTs in acid reflux is intriguing and may indicate a selection effect related to sampling quality or an unknown biological mechanism D'Souza et al. [2021].

8 Cardiovascular Disease

8.1 Introduction

The gut microbiome influences cardiovascular disease (CVD) through several mechanisms, most notably the trimethylamine N-oxide (TMAO) pathway where gut bacteria convert dietary choline and carnitine to trimethylamine, subsequently oxidized to the pro-atherogenic metabolite TMAO Wang et al. [2011], Tang et al. [2013]. Atherosclerotic CVD patients exhibit dysbiosis with increased Enterobacteriaceae and *Streptococcus* Jie et al. [2017], and TMAO directly enhances platelet reactivity and thrombosis Zhu et al. [2016]. Oral bacteria, particularly Pasteurellaceae, have been detected in atherosclerotic plaques, implicating oral–gut–cardiovascular cross-talk.

8.2 Results

8.2.1 SILVA Univariate Associations

Significant univariate DCST associations with cardiovascular disease: *Pasteurellaceae* (enriched, OR=8.77, $q = 2.0e-28$), *Staphylococcus* (enriched, OR=4.22, $q = 2.3e-17$), *Neisseria* (enriched, OR=4.51, $q = 2.7e-08$), *Chloroplast* (enriched, OR=4.70, $q = 3.4e-06$), *Prevotella 7* (depleted, OR=0.13, $q = 1.8e-03$), *Escherichia-Shigella* (depleted, OR=0.74, $q = 5.5e-03$).

8.2.2 SILVA Covariate-Adjusted Associations

No associations reached significance after covariate adjustment.

8.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Pasteurellaceae* (enriched, OR=9.50), *Staphylococcus* (enriched, OR=4.52), *JC017* (enriched, OR=6.13), *Bacteria; ; ; ; ; ;* (depleted, OR=0.74), *Acinetobacter* (enriched, OR=2.55).

8.2.4 Depth-2 Subtype Analysis

Agathobacter (adjOR=3.47, $q = 2.6e-05$).

8.2.5 Sensitivity Analysis

No associations survived the sensitivity analysis after excluding potentially contaminated samples.

8.3 Discussion

The CVD associations in our data are striking but must be interpreted cautiously: the strong enrichment of *Pasteurellaceae*-dominant DCSTs (OR=8.77) and *Staphylococcus* (OR=4.22) in univariate analysis did not survive covariate adjustment, suggesting confounding by age and other demographic factors. CVD disproportionately affects older individuals, and these taxa may correlate with age-related microbiome changes. The *Pasteurellaceae* result is nonetheless biologically plausible given the documented presence of oral bacteria in atherosclerotic lesions Jie et al. [2017]. The depth-2 *Agathobacter* enrichment (adjOR=3.47) is a novel finding warranting

further investigation, as *Agathobacter rectalis* is typically considered a beneficial SCFA producer Wang et al. [2011]. GG2 analysis confirmed the *Pasteurellaceae* and *Staphylococcus* associations univariately.

9 Obesity

9.1 Introduction

The relationship between gut microbiome and obesity has been studied extensively since landmark findings that the obese microbiome has increased energy harvest capacity Turnbaugh et al. [2006], Ley et al. [2006]. The Firmicutes/Bacteroidetes ratio was initially proposed as a biomarker, though subsequent meta-analyses questioned its consistency Walters et al. [2014]. Low microbial gene richness correlates with metabolic dysfunction Le Chatelier et al. [2013], and *Akkermansia muciniphila* has shown anti-obesity effects through enhanced barrier function Everard et al. [2013]. Enterotype classification (Bacteroides vs. Prevotella dominant) may predict dietary response in obesity Arumugam et al. [2011].

9.2 Results

9.2.1 SILVA Univariate Associations

Significant univariate DCST associations with obesity: *Prevotella 7* (enriched, OR=2.75, $q = 1.2e - 14$), *Escherichia-Shigella* (depleted, OR=0.85, $q = 4.5e - 03$), *Lactobacillus* (enriched, OR=2.46, $q = 1.9e - 02$).

9.2.2 SILVA Covariate-Adjusted Associations

No associations reached significance after covariate adjustment.

9.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Prevotella* (enriched, OR=2.52), *Bacteroides H 857956* (enriched, OR=1.86), *Bacteria; ; ; ; ; ;* (depleted, OR=0.83), *Bacteroides H 857956* (enriched, OR=1.95), *Phocaeicola A* (enriched, OR=1.24) (showing top 5 of 8).

9.2.4 Depth-2 Subtype Analysis

No depth-2 associations reached significance.

9.2.5 Sensitivity Analysis

No associations survived the sensitivity analysis after excluding potentially contaminated samples.

9.3 Discussion

Our analysis identified only univariate associations with obesity, with the *Prevotella_7*-dominant DCST showing the strongest enrichment (OR=2.75). This finding aligns with the literature on Prevotella-dominated gut types and their complex relationship with diet and obesity; high Prevotella/Bacteroidetes ratios predict greater weight loss on high-fiber diets but may also associate with higher BMI in certain dietary contexts Arumugam et al. [2011]. The lack of adjusted associations suggests that demographic confounders (age, sex) explain much of the variance, consistent with the well-documented complexity of obesity–microbiome relationships

Walters et al. [2014]. GG2 analysis identified additional signals including *Bacteroides_H* enrichment, which may represent reclassified *Bacteroides* species under updated taxonomy McDonald et al. [2024].

10 Seasonal Allergies

10.1 Introduction

The hygiene hypothesis, first articulated by Strachan Strachan [1989], posits that reduced early microbial exposure underlies the rise in allergic diseases. Modern reformulations focus on the microbiome: delayed gut microbiota maturation in infancy is a hallmark of all pediatric allergic diseases Hoskinson et al. [2023]. Analysis of the American Gut Project specifically identified reduced Clostridiales and increased Bacteroidales in allergic individuals Hua et al. [2016]. Short-chain fatty acids, particularly butyrate, protect against allergic sensitization by promoting regulatory T cells and strengthening epithelial barriers.

10.2 Results

10.2.1 SILVA Univariate Associations

Significant univariate DCST associations with seasonal allergies: *Prevotella 7* (depleted, OR=0.22, $q = 2.0e - 28$), *Escherichia-Shigella* (enriched, OR=1.21, $q = 4.4e - 09$), *Prevotella 9* (depleted, OR=0.75, $q = 1.2e - 06$), *Enterobacteriaceae* (enriched, OR=1.25, $q = 5.8e - 04$), *Pasteurellaceae* (depleted, OR=0.58, $q = 2.1e - 03$), *Neisseria* (depleted, OR=0.57, $q = 5.4e - 03$).

10.2.2 SILVA Covariate-Adjusted Associations

Table 5: Covariate-adjusted DCST associations with Seasonal Allergies (SILVA, $q < 0.05$).

DCST	adjOR	95% CI	q-value
Prevotella 9	0.67	0.59–0.76	3.3e-08
Escherichia-Shigella	1.13	1.06–1.21	5.1e-03
Enterobacteriaceae	1.21	1.08–1.36	2.7e-02

10.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Prevotella* (depleted, OR=0.12), *Bacteria; ; ; ; ; ;* (enriched, OR=1.22), *Lachnospira* (depleted, OR=0.06), *Enterobacteriaceae A 725029* (enriched, OR=1.26), *Pasteurellaceae* (depleted, OR=0.56) (showing top 5 of 9).

10.2.4 Depth-2 Subtype Analysis

Bacteroides (adjOR=1.25, $q = 1.2e - 05$), *Bacteroides* (adjOR=0.62, $q = 2.8e - 04$).

10.2.5 Sensitivity Analysis

3 association(s) survived: *Prevotella 9* (adjOR=0.67), *Escherichia-Shigella* (adjOR=1.14), *Enterobacteriaceae* (adjOR=1.22).

10.3 Discussion

The protective *Prevotella_9* association (adjOR=0.67) aligns directly with the AGP analysis by Hua et al. Hua et al. [2016], which identified Prevotella depletion in allergic subjects. The enrichment of *Escherichia–Shigella* and *Enterobacteriaceae* DCSTs in seasonal allergy subjects (both surviving adjustment and sensitivity analysis) is consistent with the framework that Proteobacteria-dominated gut communities reflect a dysbiotic state with reduced colonization resistance and impaired immune tolerance Hoskinson et al. [2023]. The GG2 analysis showed the same pattern with *Prevotella* depletion being among the strongest signals. All three adjusted associations survived sensitivity analysis, indicating these are robust findings independent of potential oral/skin contamination.

11 Diabetes

11.1 Introduction

The gut microbiome is implicated in both type 1 and type 2 diabetes through distinct but overlapping mechanisms. Landmark metagenomics studies revealed decreased butyrate producers and increased opportunistic pathogens in type 2 diabetes Qin et al. [2012], Karlsson et al. [2013]. Metabolic endotoxemia—elevated circulating bacterial LPS—initiates insulin resistance through TLR4 signaling Cani et al. [2007]. Metformin's therapeutic effects are partly mediated through enrichment of *Akkermansia muciniphila* Forslund et al. [2015]. Recent work demonstrates that dysbiotic carbohydrate fermentation directly contributes to insulin resistance Takeuchi et al. [2023].

11.2 Results

11.2.1 SILVA Univariate Associations

Significant univariate DCST associations with diabetes: *Streptococcus* (depleted, OR=0.28, $q = 8.2e - 03$), *Staphylococcus* (depleted, OR=0.24, $q = 2.9e - 02$).

11.2.2 SILVA Covariate-Adjusted Associations

No associations reached significance after covariate adjustment.

11.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Streptococcus* (depleted, OR=0.26), *Prevotella* (enriched, OR=4.59), *Bacteroides H 857956* (enriched, OR=2.01), *Staphylococcus* (depleted, OR=0.25), *Prevotella* (depleted, OR=0.00).

11.2.4 Depth-2 Subtype Analysis

Akkermansia (adjOR=4.88, $q = 2.1e - 04$).

11.2.5 Sensitivity Analysis

No associations survived the sensitivity analysis after excluding potentially contaminated samples.

11.3 Discussion

Diabetes showed relatively few DCST associations, with only univariate depletion of *Streptococcus* and *Staphylococcus* DCSTs reaching significance. At depth-2, however, *Akkermansia*-dominant subtypes showed strong enrichment (adjOR=4.88), which is paradoxical given the protective role typically attributed to *A. muciniphila* Everard et al. [2013], Forslund et al. [2015]. This may reflect reverse causation: metformin treatment enriches *Akkermansia*, and treated diabetics would have high *Akkermansia* abundance precisely because of their medication Forslund et al. [2015]. This confounding effect highlights the challenge of cross-sectional microbiome studies in medicated populations Qin et al. [2012].

12 *Clostridioides difficile* Infection (CDI)

12.1 Introduction

Clostridioides difficile infection is fundamentally a disease of dysbiosis: antibiotic-mediated disruption of the healthy microbiome eliminates colonization resistance, enabling *C. difficile* spore germination and toxin production Sehgal and Khanna [2021], Schubert et al. [2014]. CDI patients exhibit reduced diversity with enrichment of Enterococcus and depletion of Bacteroides, Lachnospiraceae, and Ruminococcaceae. Fecal microbiota transplantation achieves 80–93% cure rates by restoring these protective communities Seekatz et al. [2014]. Key protective mechanisms include secondary bile acid production by *Clostridium scindens* and resource competition by diverse anaerobic communities.

12.2 Results

12.2.1 SILVA Univariate Associations

Significant univariate DCST associations with *clostridioides difficile* infection: *Escherichia-Shigella* (enriched, OR=1.45, $q = 5.5e-03$), *Prevotella 9* (depleted, OR=0.47, $q = 3.0e-02$).

12.2.2 SILVA Covariate-Adjusted Associations

Table 6: Covariate-adjusted DCST associations with *Clostridioides difficile* Infection (CDI) (SILVA, $q < 0.05$).

DCST	adjOR	95% CI	q-value
Escherichia-Shigella	1.48	1.19–1.84	1.1e-02

12.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Bacteroides H 857956* (enriched, OR=2.81), *Streptococcus* (depleted, OR=0.21), *Proteus* (enriched, OR=5.43), *Phocaeicola A* (enriched, OR=1.93), *Pseudomonadaceae* (enriched, OR=3.34).

12.2.4 Depth-2 Subtype Analysis

Sutterella (adjOR=9.85, $q = 1.2e-05$).

12.2.5 Sensitivity Analysis

2 association(s) survived: *Escherichia-Shigella* (adjOR=1.41), *Prevotella 9* (adjOR=0.34).

12.3 Discussion

The *Escherichia–Shigella* DCST enrichment in CDI (adjOR=1.48) is consistent with the well-documented expansion of Enterobacteriaceae in CDI-susceptible gut communities Schubert et al. [2014]. This association survived both covariate adjustment and sensitivity analysis, reinforcing its robustness. The depth-2 *Sutterella* association (adjOR=9.85) is novel; while *Sutterella* has

been reported as enriched in some inflammatory conditions, its role in CDI susceptibility deserves further investigation. The protective trend for *Prevotella_9* aligns with evidence that diverse anaerobic communities provide colonization resistance against *C. difficile* Sehgal and Khanna [2021], Seekatz et al. [2014].

13 Lung Disease

13.1 Introduction

The gut–lung axis represents a bidirectional communication pathway where gut microbiota composition influences respiratory health Budden et al. [2017]. Short-chain fatty acids from gut bacterial fermentation protect against allergic airway inflammation Trompette et al. [2014], while airway *Prevotella* enhance pathogen clearance through TLR2-dependent immune activation Horn et al. [2022]. COPD patients exhibit characteristic gut dysbiosis with altered Streptococcus species and lipid metabolites correlating with reduced lung function.

13.2 Results

13.2.1 SILVA Univariate Associations

Significant univariate DCST associations with lung disease: *Prevotella 7* (depleted, OR=0.23, $q = 1.5e - 07$), *Streptococcus* (depleted, OR=0.65, $q = 3.2e - 02$), *Pasteurellaceae* (depleted, OR=0.44, $q = 4.1e - 02$).

13.2.2 SILVA Covariate-Adjusted Associations

Table 7: Covariate-adjusted DCST associations with Lung Disease (SILVA, $q < 0.05$).

DCST	adjOR	95% CI	q-value
Staphylococcus	1.55	1.17–2.05	3.9e-02

13.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Prevotella* (depleted, OR=0.23), *Streptococcus* (depleted, OR=0.61), *Pasteurellaceae* (depleted, OR=0.46).

13.2.4 Depth-2 Subtype Analysis

Agathobacter (adjOR=1.89, $q = 6.5e - 04$), *Staphylococcus* (adjOR=2.04, $q = 3.7e - 02$).

13.2.5 Sensitivity Analysis

No associations survived the sensitivity analysis after excluding potentially contaminated samples.

13.3 Discussion

Our analysis found limited but intriguing lung disease associations. The *Staphylococcus*-dominant DCST enrichment (adjOR=1.55) may reflect the known role of *S. aureus* in respiratory infections and its bidirectional translocation between gut and airways. The univariate depletion of *Prevotella_7* (OR=0.23) is consistent with the protective role of Prevotella species in respiratory immunity Horn et al. [2022]. The depth-2 *Agathobacter* enrichment (adjOR=1.89) parallels the CVD finding and warrants investigation of whether SCFA-producer dominated guts reflect

metabolic changes associated with chronic inflammatory conditions. None of the adjusted associations survived sensitivity analysis, suggesting potential confounding by oral-associated taxa Budden et al. [2017].

14 Migraine

14.1 Introduction

The gut–brain axis is increasingly implicated in migraine pathophysiology. Systematic reviews confirm altered microbiota composition in migraine patients Gorenshtein et al. [2025], with dysbiosis affecting serotonin biosynthesis, CGRP release from trigeminal neurons, and systemic inflammation. Fecal microbiota transplantation has shown promise as a durable intervention Kappéter et al. [2023]. Mendelian randomization studies have identified *Bacteroides fragilis* and *Blautia* as protective against migraine risk.

14.2 Results

14.2.1 SILVA Univariate Associations

Significant univariate DCST associations with migraine: *Bacteroides* (enriched, OR=1.19, $q = 3.7e - 04$), *Corynebacteriaceae* (depleted, OR=0.20, $q = 4.8e - 02$).

14.2.2 SILVA Covariate-Adjusted Associations

Table 8: Covariate-adjusted DCST associations with Migraine (SILVA, $q < 0.05$).

DCST	adjOR	95% CI	q-value
Bacteroides	1.16	1.05–1.26	3.4e-02

14.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Bacteroides H 857956* (enriched, OR=1.68), *Lactobacillus* (enriched, OR=2.34), *Bacteroides H 857956* (enriched, OR=1.48).

14.2.4 Depth-2 Subtype Analysis

No depth-2 associations reached significance.

14.2.5 Sensitivity Analysis

No associations survived the sensitivity analysis after excluding potentially contaminated samples.

14.3 Discussion

The modest *Bacteroides*-dominant DCST enrichment in migraine (adjOR=1.16) is consistent with recent Mendelian randomization findings and may reflect metabolite-driven mechanisms, as *Bacteroides* species produce substantial amounts of propionate, which affects vagal afferent signaling Gorenshtein et al. [2025], Cryan and Dinan [2012]. The univariate depletion of *Corynebacteriaceae* DCSTs is difficult to interpret biologically. The relatively weak associations overall suggest that DCSTs based on dominant taxa may not capture the subtle compositional shifts relevant to migraine, where functional metabolite profiles may be more informative than dominant-taxon community typing Kappéter et al. [2023].

15 Kidney Disease

15.1 Introduction

The gut–kidney axis links intestinal dysbiosis to chronic kidney disease (CKD) progression through uremic toxin production. Gut bacteria metabolize dietary precursors to produce trimethylamine N-oxide (TMAO), indoxyl sulfate, and p-cresyl sulfate, all of which accelerate renal decline Tang et al. [2015]. CKD itself alters the gut environment, creating a vicious cycle of dysbiosis and uremia Vaziri et al. [2013]. *Lactobacillus* supplementation has shown protective effects in diabetic kidney disease models.

15.2 Results

15.2.1 SILVA Univariate Associations

Significant univariate DCST associations with kidney disease: No univariate associations reached significance.

15.2.2 SILVA Covariate-Adjusted Associations

Table 9: Covariate-adjusted DCST associations with Kidney Disease (SILVA, $q < 0.05$).

DCST	adjOR	95% CI	q-value
Lactobacillus	5.85	2.26–15.14	7.8e-03
Prevotella 7	4.57	1.97–10.59	1.0e-02

15.2.3 GreenGenes2 Replication

Under GG2 taxonomy: *Bacteroides H 857956* (enriched, OR=3.85), *Prevotella* (enriched, OR=5.30).

15.2.4 Depth-2 Subtype Analysis

Lactobacillus (adjOR=5.85, $q = 1.8e - 02$).

15.2.5 Sensitivity Analysis

2 association(s) survived: *Lactobacillus* (adjOR=6.26), *Prevotella 7* (adjOR=4.55).

15.3 Discussion

The *Lactobacillus*-dominant DCST enrichment in kidney disease (adjOR=5.85) is the most noteworthy finding. While *Lactobacillus* is generally considered beneficial, elevated abundance may reflect a compensatory response to CKD-associated dysbiosis or a selection effect related to probiotic use among kidney disease patients. Alternatively, certain *Lactobacillus* species produce D-lactate, which accumulates in renal failure Vaziri et al. [2013]. The *Prevotella_7* enrichment (adjOR=4.57) deserves attention given that Prevotellaceae have been identified as CKD-protective in Mendelian randomization studies, though elevated abundance in our

cross-sectional data may again reflect reverse causation. Both associations survived sensitivity analysis, supporting their robustness Tang et al. [2015].

16 General Discussion

This Phase 1 analysis demonstrates that DCST classification—a simple, clustering-free method of community typing based on the dominant taxon after L_∞ normalization—identifies biologically meaningful associations between gut microbiota structure and diverse health outcomes. Across 12 conditions, we identified 28 covariate-adjusted associations (SILVA) and 23 that survived sensitivity analysis, indicating that the majority of findings are robust to potential oral/skin contamination artifacts.

Several cross-cutting patterns emerge. First, *Bacteroides*-dominant guts show consistent positive associations with inflammatory conditions (IBD, autoimmune disease, acid reflux, migraine), suggesting that Bacteroides dominance may mark a community state with reduced resilience to inflammatory challenges. Second, *Prevotella*-dominant states are consistently protective against allergies and autoimmune conditions, consistent with the hygiene hypothesis framework and the ecological trade-off between Prevotella and Bacteroides community types Arumugam et al. [2011]. Third, *Morganella* dominance—a rare but strongly disease-associated DCST—emerges as a powerful marker for IBD and autoimmune conditions, likely reflecting severe dysbiosis with expansion of opportunistic Enterobacteriaceae.

The GG2 analysis provides independent taxonomic validation, with 91.6% genus-level concordance and qualitatively similar association patterns. This dual-taxonomy approach strengthens confidence in our findings and demonstrates the robustness of DCST classification to reference database choice.

16.1 Limitations

Key limitations include: (1) cross-sectional design precludes causal inference; (2) self-reported health conditions in AGP are not clinically verified; (3) potential confounding by diet, medication, and geography not captured in available metadata; (4) DCSTs capture only the dominant taxon and may miss more subtle compositional shifts; (5) Obesity was derived from BMI ≥ 30 rather than clinical diagnosis. Despite these limitations, the large sample size, consistent results across taxonomic schemas, and robustness to sensitivity analysis support the utility of DCSTs as a practical tool for community-level microbiome epidemiology.

16.2 Conclusions

DCST classification provides a computationally simple yet informative framework for characterizing gut microbiota community structure at scale. The method identifies disease associations that are biologically plausible, reproducible across taxonomic references, and robust to quality-control sensitivity analysis. Future work should extend this approach to longitudinal cohorts with clinical phenotyping and integrate functional metagenomics data to elucidate mechanistic pathways underlying DCST–disease associations.

References

Manimozhiyan Arumugam, Jeroen Raes, Eric Pelletier, et al. Enterotypes of the human gut microbiome. *Nature*, 473:174–180, 2011. doi: 10.1038/nature09944.

Koji Atarashi, Taira Tanoue, Keiji Oshima, et al. Treg induction by a rationally selected mixture of Clostridia strains from the human microbiota. *Nature*, 500:232–236, 2013. doi: 10.1038/nature12331.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society Series B*, 57: 289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

Kurtis F Budden, Shaan L Gellatly, David L A Wood, et al. Emerging pathogenic links between microbiota and the gut–lung axis. *Nature Reviews Microbiology*, 15:55–63, 2017. doi: 10.1038/nrmicro.2016.142.

Patrice D Cani, Jacques Amar, Miguel A Iglesias, et al. Metabolic endotoxemia initiates obesity and insulin resistance. *Diabetes*, 56:1761–1772, 2007. doi: 10.2337/db06-1491.

John F Cryan and Timothy G Dinan. Mind-altering microorganisms: The impact of the gut microbiota on brain and behaviour. *Nature Reviews Neuroscience*, 13:701–712, 2012. doi: 10.1038/nrn3346.

F De Luca and Y Shoenfeld. The microbiome in autoimmune diseases. *Clinical and Experimental Immunology*, 195:74–85, 2019. doi: 10.1111/cei.13158.

N P Deshpande, S M Riordan, N Castaño-Rodríguez, M R Wilkins, and N O Kaakoush. Signatures within the esophageal microbiome are associated with host genetics, age, and disease. *Microbiome*, 6:227, 2018. doi: 10.1186/s40168-018-0611-4.

Timothy G Dinan and John F Cryan. Irritable bowel syndrome: A microbiome-gut-brain axis disorder? *Neurogastroenterology and Motility*, 27:445–455, 2015. doi: 10.1111/nmo.12520.

S M D'Souza, K Houston, L Keenan, B S Yoo, P J Parekh, and D A Johnson. Role of microbial dysbiosis in the pathogenesis of esophageal mucosal disease: A paradigm shift from acid to bacteria? *World Journal of Gastroenterology*, 27:2054–2072, 2021. doi: 10.3748/wjg.v27.i18.2054.

Amandine Everard, Clara Belzer, Lucie Geurts, et al. Cross-talk between *Akkermansia muciniphila* and intestinal epithelium controls diet-induced obesity. *Proceedings of the National Academy of Sciences*, 110:9066–9071, 2013. doi: 10.1073/pnas.1219451110.

Kristoffer Forslund, Falk Hildebrand, Trine Nielsen, et al. Disentangling type 2 diabetes and metformin treatment signatures in the human gut microbiota. *Nature*, 528:262–266, 2015. doi: 10.1038/nature15766.

Daniel N Frank, April L St Amand, Robert A Feldman, et al. Molecular-phylogenetic characterization of microbial community imbalances in human inflammatory bowel diseases. *Proceedings of the National Academy of Sciences*, 104:13780–13785, 2007. doi: 10.1073/pnas.0706625104.

Pawel Gajer, Rebecca M Brotman, Guoyun Bai, et al. Temporal dynamics of the human vaginal microbiota. *Science Translational Medicine*, 4:132ra52, 2012. doi: 10.1126/scitranslmed.3003605.

Dirk Gevers, Subra Kugathasan, Lee A Denson, et al. The treatment-naive microbiome in new-onset Crohn's disease. *Cell Host and Microbe*, 15:382–392, 2014. doi: 10.1016/j.chom.2014.02.005.

Alon Gorenshtein, Kamel Shihada, Liron Leibovitch, Tom Liba, and Avner Goren. The association between migraine and gut microbiota: a systematic review. *Acta Neurologica Belgica*, 125:977–987, 2025. doi: 10.1007/s13760-025-02779-y.

Kadi J Horn, Melissa A Schopper, Zoe G Drigot, and Sarah E Clark. Airway *Prevotella* promote TLR2-dependent neutrophil activation and rapid clearance of *Streptococcus pneumoniae* from the lung. *Nature Communications*, 13:3321, 2022. doi: 10.1038/s41467-022-31074-0.

C L Hoskinson, D L Y Dai, K L Del Bel, et al. Delayed gut microbiota maturation in the first year of life is a hallmark of pediatric allergic disease. *Nature Communications*, 14:4785, 2023. doi: 10.1038/s41467-023-40336-4.

Xinmin Hua, James J Goedert, Angela Pu, Guoqin Yu, and Jianxin Shi. Allergy associations with the adult fecal microbiota: Analysis of the American Gut Project. *EBioMedicine*, 3:172–179, 2016. doi: 10.1016/j.ebiom.2015.11.038.

Zhuye Jie, Huihua Xia, Shi-Long Zhong, et al. The gut microbiome in atherosclerotic cardiovascular disease. *Nature Communications*, 8:845, 2017. doi: 10.1038/s41467-017-00900-1.

Ágnes Kappéter, Dávid Sipos, Adorján Varga, et al. Migraine as a disease associated with dysbiosis and possible therapy with fecal microbiota transplantation. *Microorganisms*, 11:2083, 2023. doi: 10.3390/microorganisms11082083.

Fredrik H Karlsson, Valentina Tremaroli, Intawat Nookaew, et al. Gut metagenomics in European women with normal, impaired and diabetic glucose control. *Nature*, 498:99–103, 2013. doi: 10.1038/nature12198.

Emmanuelle Le Chatelier, Trine Nielsen, Junjie Qin, et al. Richness of human gut microbiome correlates with metabolic markers. *Nature*, 500:541–546, 2013. doi: 10.1038/nature12506.

Ruth E Ley, Peter J Turnbaugh, Samuel Klein, and Jeffrey I Gordon. Human gut microbes associated with obesity. *Nature*, 444:1022–1023, 2006. doi: 10.1038/4441022a.

Jason Lloyd-Price, Cesar Arze, Ashwin N Ananthakrishnan, et al. Multi-omics of the gut microbial ecosystem in inflammatory bowel diseases. *Nature*, 569:655–662, 2019. doi: 10.1038/s41586-019-1237-9.

Chaysip Manichanh, Laurence Rigottier-Gois, Emmanuelle Bonnaud, et al. Reduced diversity of faecal microbiota in Crohn's disease revealed by a metagenomic approach. *Gut*, 55:205–211, 2006. doi: 10.1136/gut.2005.073817.

Daniel McDonald, Embriette Hyde, Justine W Debelius, et al. American Gut: an open platform for citizen science microbiome research. *mSystems*, 3:e00031–18, 2018. doi: 10.1128/mSystems.00031-18.

Daniel McDonald, Yueyu Jiang, Metin Balaban, et al. GreenGenes2 unifies microbial data in a single reference tree. *Nature Biotechnology*, 42:715–718, 2024. doi: 10.1038/s41587-023-02009-5.

Rapat Pittayanon, Jennifer T Lau, Yuhong Yuan, Grigorios I Leontiadis, Frances Tse, Michael G Surette, and Paul Moayyedi. Gut microbiota in patients with irritable bowel syndrome: A systematic review. *Gastroenterology*, 157:97–108, 2019. doi: 10.1053/j.gastro.2019.03.049.

Junjie Qin, Yingrui Li, Zhiming Cai, et al. A metagenome-wide association study of gut microbiota in type 2 diabetes. *Nature*, 490:55–60, 2012. doi: 10.1038/nature11450.

Christian Quast, Elmar Pruesse, Pelin Yilmaz, et al. The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. *Nucleic Acids Research*, 41:D590–D596, 2013. doi: 10.1093/nar/gks1219.

R Balfour Sartor. Microbial influences in inflammatory bowel diseases. *Gastroenterology*, 134:577–594, 2008. doi: 10.1053/j.gastro.2007.11.059.

Jose U Scher, Andrew Sczesnak, Randy S Longman, et al. Expansion of intestinal *Prevotella copri* correlates with enhanced susceptibility to arthritis. *eLife*, 2:e01202, 2013. doi: 10.7554/eLife.01202.

Allison M Schubert, Matthew A M Rogers, Corrie Ring, et al. Microbiome data distinguish patients with *Clostridium difficile* infection and non-*C. difficile*-associated diarrhea from healthy controls. *mBio*, 5:e01021–14, 2014. doi: 10.1128/mBio.01021-14.

Anna M Seekatz, Jorn Aas, Christine E Gessert, et al. Recovery of the gut microbiome following fecal microbiota transplantation. *mBio*, 5:e00893–14, 2014. doi: 10.1128/mBio.00893-14.

Kanika Sehgal and Sahil Khanna. Gut microbiome and *Clostridioides difficile* infection: a closer look at the microscopic interface. *Therapeutic Advances in Gastroenterology*, 14:1756284821994736, 2021. doi: 10.1177/1756284821994736.

Harry Sokol, Bernardo Pigneur, Laetitia Watterlot, et al. *Faecalibacterium prausnitzii* is an anti-inflammatory commensal bacterium identified by gut microbiota analysis of Crohn disease patients. *Proceedings of the National Academy of Sciences*, 105:16731–16736, 2008. doi: 10.1073/pnas.0804812105.

D P Strachan. Hay fever, hygiene, and household size. *BMJ*, 299:1259–1260, 1989. doi: 10.1136/bmj.299.6710.1259.

Takeshi Takeuchi, Tetsuya Kubota, Yumiko Nakanishi, et al. Gut microbial carbohydrate metabolism contributes to insulin resistance. *Nature*, 621:591–598, 2023. doi: 10.1038/s41586-023-06466-x.

W H Wilson Tang, Zeneng Wang, Bruce S Levison, et al. Intestinal microbial metabolism of phosphatidylcholine and cardiovascular risk. *New England Journal of Medicine*, 368:1575–1584, 2013. doi: 10.1056/NEJMoa1109400.

W H Wilson Tang, Zeneng Wang, David J Kennedy, et al. Gut microbiota-dependent trimethylamine N-oxide (TMAO) pathway contributes to both development of renal insufficiency and mortality risk in chronic kidney disease. *Circulation Research*, 116:448–455, 2015. doi: 10.1161/CIRCRESAHA.116.305360.

Julien Tap, Muriel Derrien, Henrik Törnblom, Romain Brazeilles, Sonia Cools-Portier, Joël Doré, Sanna Störsrud, Benjamin Le Nevé, and Magnus Simrén. Identification of an intestinal microbiota signature associated with severity of irritable bowel syndrome. *Gastroenterology*, 152:111–123, 2017. doi: 10.1053/j.gastro.2016.09.049.

Aurélien Trompette, Evan S Gollwitzer, Khyati Yadava, et al. Gut microbiota metabolism of dietary fiber influences allergic airway disease and hematopoiesis. *Nature Medicine*, 20:159–166, 2014. doi: 10.1038/nm.3444.

Peter J Turnbaugh, Ruth E Ley, Michael A Mahowald, Vincent Magrini, Elaine R Mardis, and Jeffrey I Gordon. An obesity-associated gut microbiome with increased capacity for energy harvest. *Nature*, 444:1027–1031, 2006. doi: 10.1038/nature05414.

Tommi Vatanen, Eric A Franzosa, Randall Schwager, et al. The human gut microbiome in early-onset type 1 diabetes from the TEDDY study. *Nature*, 562:589–594, 2018. doi: 10.1038/s41586-018-0620-2.

Nosratola D Vaziri, Jakk Wong, Madeleine Pahl, et al. Chronic kidney disease alters intestinal microbial flora. *Kidney International*, 83:308–315, 2013. doi: 10.1038/ki.2012.345.

William A Walters, Zech Xu, and Rob Knight. Meta-analyses of human gut microbes associated with obesity and IBD. *FEBS Letters*, 588:4223–4233, 2014. doi: 10.1016/j.febslet.2014.09.039.

K Wang, S Wang, Y Chen, X Lu, D Wang, Y Zhang, W Pan, C Zhou, and D Zou. Causal relationship between gut microbiota and risk of gastroesophageal reflux disease: a genetic correlation and bidirectional Mendelian randomization study. *Frontiers in Immunology*, 15:1327503, 2024. doi: 10.3389/fimmu.2024.1327503.

Zeneng Wang, Erin Klipfell, Brian J Bennett, et al. Gut flora metabolism of phosphatidylcholine promotes cardiovascular disease. *Nature*, 472:57–63, 2011. doi: 10.1038/nature09922.

W Zhu, J C Gregory, E Org, et al. Gut microbe-generated trimethylamine N-oxide from dietary choline is prothrombotic in subjects. *Cell*, 165:111–124, 2016. doi: 10.1016/j.cell.2016.02.011.